Supplementary Material for "Integrated-Haar wavelet collocation with a provably real spectrum on arbitrary grids"

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Part A states the results of the main text formally and proves them. Part B contains additional tables and a figure, all generated by the scripts of the repository https://github.com/mossaiby/iHaar.

Part A. Statements and proofs

Notation

The grid is $a = t_0 < t_1 < \cdots < t_N = b$, with cells $I_k = [t_{k-1}, t_k]$ of width h_k, midpoints m_k and midpoint spacings $d_k = (h_k + h_{k+1})/2$. The trial space $\mathcal{S}_h$ consists of the C^1 quadratic splines on the grid, $U_k = s(m_k)$ are the nodal values, and $c_k = s''|_{I_k}$ are the cell values of the second derivative. The second-derivative operator is $D_2 = A^{-1}L$, and δ^2 is the three-point second-difference matrix.

Statements

Result 1 (Three-term identity and solvability)**.** *For every $s \in \mathcal{S}_h$, with $U_k = s(m_k)$ and $c_k = s''|_{I_k}$, and for $2 \le k \le N-1$,*

$$\frac{U_{k+1} - U_k}{d_k} - \frac{U_k - U_{k-1}}{d_{k-1}} = \frac{h_{k-1}^2}{8d_{k-1}} c_{k-1} + \Big(h_k - \frac{h_k^2}{8d_{k-1}} - \frac{h_k^2}{8d_k}\Big) c_k + \frac{h_{k+1}^2}{8d_k} c_{k+1}. \tag{1}$$

At the left end, with $g_0 = s(a)$,

$$\frac{U_2 - U_1}{d_1} - \frac{2(U_1 - g_0)}{h_1} = \Big(\frac{3h_1}{4} - \frac{h_1^2}{8d_1}\Big) c_1 + \frac{h_2^2}{8d_1} c_2 \qquad \textit{(value } g_0 \textit{ prescribed)}, \tag{2}$$

$$\frac{U_2 - U_1}{d_1} = \Big(h_1 - \frac{h_1^2}{8d_1}\Big) c_1 + \frac{h_2^2}{8d_1} c_2 \qquad \textit{(if } s'(a) = 0\textit{)}, \tag{3}$$

and symmetrically at the right end. In matrix form, $LU + b_0 g_0 + b_1 g_1 = Ac$, where L is a tridiagonal difference matrix and A a tridiagonal averaging matrix. Moreover, on every grid, A has a positive diagonal, positive off-diagonal entries, and is strictly diagonally dominant by columns; hence it is invertible. With values prescribed at both ends, the collocation problem $s''(m_k) = f(m_k)$ has a unique solution.

Result 2 (Spectrum on every grid)**.** *Let $a = t_0 < t_1 < \cdots < t_N = b$, with $N \geq 2$, be any strictly increasing knot vector. At each end, let either the value or the slope (zero slope) be prescribed. Then $D_2 = A^{-1}L$ has N real and simple eigenvalues, and it is diagonalizable with a real eigenvector matrix V. If a value is prescribed at one or both ends, all eigenvalues are negative. If the slope vanishes at both ends, exactly one eigenvalue is zero, with the constant vector as eigenvector, and all others are negative.*

Result 3 (Uniform grids and fourth-order correction)**.** *On a uniform grid ($h_k \equiv h$), D_2 is symmetric,*

$$D_2 = \frac{8}{h^2}\,\delta^2\,(8I + \delta^2)^{-1}, \tag{4}$$

where δ^2 is the usual three-point second-difference matrix, with first and last diagonal entry -3 for prescribed values and -1 for zero slope. Its eigenvectors are the DST-II (prescribed values) or DCT-II (zero slope) vectors, and its eigenvalues $\lambda_k = 8\mu_k/(h^2(8 + \mu_k))$, with $\mu_k = -4\sin^2(\pi k/2N)$, lie in $[-8/h^2, 0]$. Moreover, on a uniform grid, for smooth u satisfying zero-slope conditions with all odd normal derivatives vanishing, or periodic conditions,

$$D_2U = u''(m) - \tfrac{h^2}{24}u^{(4)}(m) + O(h^4), \qquad D_2^{(4)}U = u''(m) + O(h^4), \qquad D_2^{(4)} := D_2 + \tfrac{h^2}{24}D_2^2. \tag{5}$$

$D_2^{(4)}$ has the same eigenvectors as D_2 and the eigenvalues $\lambda_k + h^2\lambda_k^2/24 \in [-\frac{16}{3h^2}, 0]$, so it is also symmetric and negative. The same holds for prescribed values when the even derivatives vanish at the ends.

Result 4 (Exact mass conservation)**.** *With zero-slope ends, on any grid, $h^{\mathsf{T}}D_2 = 0$ with $h = (h_k)$. On uniform grids, moreover, $h\sum_k U_k$ equals the exact integral of the spline.*

For the phase-field schemes on a uniform grid, let $K = -\Delta_h$ be the negative of the tensor-sum discrete Laplacian built from D_2 or $D_2^{(4)}$, $\langle v, w\rangle_h = h^d \sum vw$, $E_{1,h}(u) = \langle \frac{1}{4}(u^2 - 1)^2, 1\rangle_h$, $f(u) = u^3 - u$, $G_h = I$ (Allen–Cahn) or $G_h = K$ (Cahn–Hilliard), $\bar{u} = 2u^n - u^{n-1}$ and $\bar{b} = f(\bar{u})/\sqrt{E_{1,h}(\bar{u}) + C}$ with $C > 0$, and $S \geq 0$. The stabilized SAV-BDF2 scheme is

$$\frac{3u^{n+1} - 4u^n + u^{n-1}}{2\Delta t} = -G_h\mu^{n+1}, \qquad \mu^{n+1} = \varepsilon^2 K u^{n+1} + S(u^{n+1} - \bar{u}) + r^{n+1}\bar{b}, \tag{6}$$

$$3r^{n+1} - 4r^n + r^{n-1} = \tfrac{1}{2}\langle \bar{b}, 3u^{n+1} - 4u^n + u^{n-1}\rangle_h, \tag{7}$$

Result 5 (Solvability, mass and modified-energy decay)**.** *On a uniform grid, for every $\Delta t > 0$ and $S \geq 0$, the scheme has a unique solution. For CH, the discrete mass $\langle u^n, 1\rangle_h$, which equals the exact integral of the spline, is constant. The modified energy*

$$\tilde{E}^{n+1} = \tfrac{\varepsilon^2}{4}\big(\|u^{n+1}\|_K^2 + \|2u^{n+1} - u^n\|_K^2\big) + \tfrac{1}{2}\big((r^{n+1})^2 + (2r^{n+1} - r^n)^2\big) + \tfrac{S}{2}\|u^{n+1} - u^n\|_h^2$$

never increases: $\tilde{E}^{n+1} \leq \tilde{E}^n$. This holds for both D_2 and $D_2^{(4)}$.

Proofs

Result 1

Proof of Result 1. Let $\sigma_k = s'(t_k)$. On I_k, $s'(x) = \sigma_k + c_k(x - t_k) = \sigma_{k-1} + c_k(x - t_{k-1})$, hence $\sigma_k - \sigma_{k-1} = c_k h_k$. Integrating s' from m_k to m_{k+1} across t_k gives

$$U_{k+1} - U_k = \int_{m_k}^{t_k} \big(\sigma_k + c_k(x - t_k)\big)\,dx + \int_{t_k}^{m_{k+1}} \big(\sigma_k + c_{k+1}(x - t_k)\big)\,dx = \sigma_k d_k - \frac{c_k h_k^2}{8} + \frac{c_{k+1} h_{k+1}^2}{8}.$$

Dividing by d_k, subtracting the same relation for $k-1$, and using $\sigma_k - \sigma_{k-1} = c_k h_k$ gives (1). At the left end, $U_1 - g_0 = \sigma_0 h_1/2 + c_1 h_1^2/8$, so $2(U_1 - g_0)/h_1 = \sigma_0 + c_1 h_1/4$; subtracting this from the relation for $(U_2 - U_1)/d_1$ gives (2). If $\sigma_0 = 0$, then $\sigma_1 = c_1 h_1$ and (3) follows. The right end is analogous. The expressions for $s'(m_k)$ and for the end slopes follow from $s' = \sigma_0 + \int s''$. □

Proof of Result 1. The off-diagonal entries of column k of A are positive and sum to $h_k^2/(8d_{k-1}) + h_k^2/(8d_k)$, so the column-dominance margin is $h_k - h_k^2/(4d_{k-1}) - h_k^2/(4d_k) > 0$, because $d_j > h_k/2$. For an end column with a prescribed value the margin is $3h_1/4 - h_1^2/(4d_1) > h_1/4$, and for a zero-slope end column it is $h_1 - h_1^2/(4d_1) > h_1/2$. With values prescribed at both ends, L is a symmetric negative definite three-point matrix, so $LU = Af - b_0 g_0 - b_1 g_1$ has a unique solution. □

Result 2

Proof. We use three standard facts [1, 2]. (F1) A tridiagonal matrix with nonnegative entries is totally nonnegative (all minors nonnegative) if and only if all its principal minors are nonnegative. (F2) If B is nonsingular and totally nonnegative, then $JB^{-1}J$ is totally nonnegative, where $J = \mathrm{diag}((-1)^i)$; products of totally nonnegative matrices are totally nonnegative (Cauchy–Binet). (F3) A totally nonnegative matrix is oscillatory if it is nonsingular and its entries next to the diagonal are positive; the eigenvalues of an oscillatory matrix are real, positive and simple (Gantmacher–Krein).

First let a value be prescribed at one or both ends. Then $-L$ is an irreducible tridiagonal matrix with positive diagonal, negative off-diagonal entries, zero row sums in the interior and a positive row sum in at least one end row. It is therefore a nonsingular M-matrix, and all its principal minors are positive. The matrix $J(-L)J$ has the same principal minors and nonnegative entries, so it is totally nonnegative by (F1). By (F2), $(-L)^{-1} = J\,(J(-L)J)^{-1}J$ is totally nonnegative; it is also entrywise positive, as the inverse of an irreducible M-matrix. The matrix A has nonnegative entries, and JAJ is a Z-matrix that is strictly diagonally dominant by columns (Result 1), hence an M-matrix. Its principal minors, which are those of A, are therefore positive, and A is totally nonnegative by (F1). The product $P = (-L)^{-1}A$ is totally nonnegative and nonsingular, and it is entrywise positive, since $(-L)^{-1}$ is entrywise positive and A is nonnegative with a positive diagonal. By (F3), P is oscillatory and has N real, positive, simple eigenvalues. Since $D_2 = A^{-1}L = -P^{-1}$, the eigenvalues of D_2 are real, negative and simple.

If the slope vanishes at both ends, L is singular. Let $0 < \sigma < 8/\max_k h_k^2$, and consider $-(L - \sigma A) = -L + \sigma A$. Its off-diagonal entries are $-1/d_k + \sigma h^2/(8d_k) < 0$, and its columns have positive dominance margin σh_k, because the columns of L sum to zero and those of

A sum to h_k (Result 4). It is therefore a nonsingular M-matrix, and the argument above applies to $(-L+\sigma A)^{-1}A$. Hence $A^{-1}L-\sigma I$ has real, negative, simple eigenvalues, so the eigenvalues of D_2 are real, simple and smaller than σ. Since σ can be arbitrarily small, they are nonpositive, and the constant vector spans the kernel. □

Result 3: symmetry and spectrum

Proof. With $h_k = h$ we have $d_k = h$, and the coefficients of (1)–(3) become $h(\frac{1}{8}, \frac{6}{8}, \frac{1}{8})$, $h(\frac{5}{8}, \frac{1}{8})$ and $h(\frac{7}{8}, \frac{1}{8})$. These are the rows of $h(I + \delta^2/8)$, while the left-hand sides are the rows of δ^2/h. Since $I+\delta^2/8$ is a polynomial in the symmetric matrix δ^2, the two commute, and $D_2 = h^{-2}(I + \delta^2/8)^{-1}\delta^2$ is symmetric. The end rows of δ^2 correspond to ghost values $U_0 = -U_1$ (odd reflection) or $U_0 = U_1$ (even reflection), whose eigenvectors are the DST-II and DCT-II vectors with eigenvalues $\mu_k \in [-4, 0]$. Finally, $\mu \mapsto 8\mu/(8 + \mu)$ maps $[-4, 0]$ monotonically onto $[-8, 0]$. □

Result 3: fourth-order correction

Proof. By the reflection property, the zero-slope matrices act on U in the same way as the circulant matrices of the $2N$-point periodic grid act on the even extension. For a sampled exponential $e^{i\omega x}$, the circulant D_2 has the eigenvalue $\lambda(\omega h)/h^2$, with the 2π-periodic analytic symbol $\lambda(\theta) = 8\mu/(8+\mu)$, where $\mu = -4\sin^2(\theta/2) = -\theta^2+\theta^4/12+O(\theta^6)$. Hence $\lambda = \mu - \mu^2/8 + O(\mu^3) = -\theta^2 - \theta^4/24 + O(\theta^6)$, which gives the first statement mode by mode. The symbol of $D_2^{(4)}$ is $\lambda + \lambda^2/24 = -\theta^2 + O(\theta^6)$, so $|\lambda_4(\omega h)/h^2 + \omega^2| \le Ch^4\omega^6$ for $|\omega h| \le \pi$. For $|\omega h| > \pi$ the bound holds trivially, because $|\lambda_4| \le 16/3$. Summing over the rapidly decaying Fourier coefficients of the smooth periodic extension gives the $O(h^4)$ bound. The spectral statement follows because $\lambda+h^2\lambda^2/24 \le 0$ for $\lambda \in [-24/h^2, 0]$. Prescribed values with odd extensions are treated in the same way. □

Result 4

Proof. From (1) and (3), column k of A sums to $h_k - \frac{h_k^2}{8d_{k-1}} - \frac{h_k^2}{8d_k} + \frac{h_k^2}{8d_{k-1}} + \frac{h_k^2}{8d_k} = h_k$, that is, $\mathbf{1}^\mathsf{T}A = h^\mathsf{T}$. Hence $h^\mathsf{T}D_2 = \mathbf{1}^\mathsf{T}L = 0$, because the columns of the zero-slope difference matrix sum to zero. On uniform grids, $\int_{I_k} s = hU_k + h^3c_k/24$ exactly, and $\sum_k h\, c_k = s'(b) - s'(a) = 0$. □

Result 5

Proof. For solvability, eliminating μ gives $Mu^{n+1} = g + 2\Delta t\, S\, G_h\bar{u} - 2\Delta t\, r^{n+1}G_h\bar{b}$ with $M = 3I + 2\Delta t\, G_h(\varepsilon^2 K + SI)$. The matrix M is symmetric positive definite, because G_h and K commute and are positive semidefinite. Writing $u^{n+1} = p + r^{n+1}q$ and substituting into (7) gives a scalar equation for r^{n+1} whose coefficient is $3 - \frac{3}{2}\langle\bar{b}, q\rangle_h \ge 3$, since $\langle\bar{b}, q\rangle_h = -2\Delta t\langle\bar{b}, M^{-1}G_h\bar{b}\rangle_h \le 0$.

For mass conservation in CH, $\langle K\mu, 1\rangle_h = \langle\mu, K1\rangle_h = 0$, so $\langle 3u^{n+1}-4u^n+u^{n-1}, 1\rangle_h = 0$. Since the BDF1 first step also conserves mass, induction gives conservation, and Result 4 identifies the discrete sum with the integral.

For the energy, multiply (6) by $2\Delta t$ and take the inner product with μ^{n+1}:

$$\langle 3a - 4b + c, \mu^{n+1}\rangle_h = -2\Delta t \langle G_h \mu^{n+1}, \mu^{n+1}\rangle_h \le 0.$$

On the left-hand side, insert the expression for μ^{n+1} and write $a = u^{n+1}$, $b = u^n$ and $c = u^{n-1}$, and use the identities

$$\begin{aligned} 2\langle 3a - 4b + c, Ka\rangle_h &= \|a\|_K^2 - \|b\|_K^2 + \|2a - b\|_K^2 - \|2b - c\|_K^2 + \|a - 2b + c\|_K^2, \\ \langle 3a - 4b + c, a - 2b + c\rangle_h &= \|a - b\|_h^2 - \|b - c\|_h^2 + 2\|a - 2b + c\|_h^2, \end{aligned}$$

the second of which follows from $(3x - y)(x - y) = x^2 - y^2 + 2(x - y)^2$ with $x = a - b$ and $y = b - c$, together with the scalar version of the first identity applied to (7) multiplied by $2r^{n+1}$, which gives $r^{n+1}\langle \bar{b}, 3a - 4b + c\rangle_h = (r^{n+1})^2 - (r^n)^2 + (2r^{n+1} - r^n)^2 - (2r^n - r^{n-1})^2 + (r^{n+1} - 2r^n + r^{n-1})^2$. Collecting terms, the left-hand side equals $2(\tilde{E}^{n+1} - \tilde{E}^n) + \frac{\varepsilon^2}{2}\|a - 2b + c\|_K^2 + (r^{n+1} - 2r^n + r^{n-1})^2 + 2S\|a - 2b + c\|_h^2$. Equating with $-2\Delta t\langle G_h\mu^{n+1}, \mu^{n+1}\rangle_h$ and dividing by two gives

$$\begin{aligned} \tilde{E}^{n+1} - \tilde{E}^n = -\Delta t \langle G_h \mu^{n+1}, \mu^{n+1}\rangle_h - \tfrac{\varepsilon^2}{4}\|a - 2b + c\|_K^2 \\ - \tfrac{1}{2}(r^{n+1} - 2r^n + r^{n-1})^2 - S\|a - 2b + c\|_h^2 \le 0. \end{aligned}$$

The same argument applies to BDF1 and to both D_2 and $D_2^{(4)}$, since it uses only the symmetry and semidefiniteness of K in $\langle\cdot,\cdot\rangle_h$ (Result 3 and Result 3). $\square$

Part B. Additional tables and figure

Table S1: Dependence on Poisson's ratio, prescribed displacements on all edges (smooth solution).

	GMRES iterations			e_{H^1}		
ν	$N = 32$	64	128	$N = 32$	64	128
0.3	17	17	16	5.78×10^{-4}	1.45×10^{-4}	3.61×10^{-5}
0.4	23	22	22	7.63×10^{-4}	1.91×10^{-4}	4.78×10^{-5}
0.45	30	30	29	1.17×10^{-3}	2.95×10^{-4}	7.39×10^{-5}
0.49	52	57	56	4.73×10^{-3}	1.23×10^{-3}	3.11×10^{-4}
0.499	96	124	141	3.14×10^{-2}	1.07×10^{-2}	2.96×10^{-3}

Table S2: Traction on two adjacent edges: dependence on Poisson's ratio.

	GMRES iterations			e_{H^1}		
ν	$N = 32$	64	128	$N = 32$	64	128
0.3	27	27	27	1.08×10^{-3}	2.70×10^{-4}	6.75×10^{-5}
0.4	33	33	33	1.93×10^{-3}	4.84×10^{-4}	1.21×10^{-4}
0.45	44	44	44	3.69×10^{-3}	9.29×10^{-4}	2.33×10^{-4}
0.49	102	98	98	1.75×10^{-2}	4.49×10^{-3}	1.13×10^{-3}
0.499	317	377	406	1.37×10^{-1}	4.15×10^{-2}	1.10×10^{-2}

Table S3: One-dimensional model $u'' = f$ with $u(0)$ and $u'(1)$ given: maximum error with an enhanced end row of weight α.

α	$N = 32$	64	128	256	512	rate
$-1/12$	4.0×10^{-4}	1.0×10^{-4}	2.6×10^{-5}	6.5×10^{-6}	1.6×10^{-6}	2.00
0 (plain)	1.2×10^{-3}	3.0×10^{-4}	7.6×10^{-5}	1.9×10^{-5}	4.8×10^{-6}	2.00
1/4	5.4×10^{-3}	1.4×10^{-3}	3.4×10^{-4}	8.6×10^{-5}	2.2×10^{-5}	2.00
1/2	9.5×10^{-3}	2.4×10^{-3}	6.1×10^{-4}	1.5×10^{-4}	3.8×10^{-5}	2.00

Table S4: Spectrum of D_2 on random graded grids: largest relative imaginary part of the eigenvalues and largest eigenvector condition number.

ends	ratio bound	N	grids	max $\lvert\mathrm{Im}\,\lambda\rvert/\lvert\lambda\rvert$	max $\kappa(V)$
values	1.5	16	1000	0.0e+00	3.1
values	1.5	64	400	0.0e+00	8.8
values	1.5	256	60	0.0e+00	20.2
values	2	16	1000	0.0e+00	4.3
values	2	64	400	0.0e+00	11.8
values	2	256	60	0.0e+00	31.4
values	4	16	1000	0.0e+00	6.4
values	4	64	400	0.0e+00	15.4
values	4	256	60	0.0e+00	43.0
zero slope	1.5	16	1000	0.0e+00	4.4
zero slope	1.5	64	400	0.0e+00	14.8
zero slope	1.5	256	60	0.0e+00	25.6
zero slope	2	16	1000	0.0e+00	7.2
zero slope	2	64	400	0.0e+00	22.2
zero slope	2	256	60	0.0e+00	54.8
zero slope	4	16	1000	0.0e+00	10.9
zero slope	4	64	400	0.0e+00	35.2
zero slope	4	256	60	0.0e+00	100.3

Table S5: Planar interface ($\varepsilon = 0.01$): maximum nodal error for three operators.

N	nodes per $\sqrt{2}\varepsilon$	3-point FD	D_2	$D_2^{(4)}$
48	0.68	1.47×10^{-1}	4.10×10^{-2}	9.77×10^{-2}
64	0.91	1.21×10^{-1}	1.29×10^{-2}	6.29×10^{-2}
96	1.36	4.07×10^{-2}	7.32×10^{-3}	1.24×10^{-2}
128	1.81	1.54×10^{-2}	6.45×10^{-3}	2.89×10^{-3}
192	2.72	8.02×10^{-3}	3.48×10^{-3}	4.88×10^{-4}
256	3.62	4.20×10^{-3}	1.99×10^{-3}	1.66×10^{-4}
384	5.43	1.90×10^{-3}	9.23×10^{-4}	3.16×10^{-5}
512	7.24	1.07×10^{-3}	5.28×10^{-4}	9.78×10^{-6}

Table S6: Cahn–Hilliard: temporal convergence of SAV-BDF2 ($N = 128$, $D_2^{(4)}$).

Δt	error	rate
1.60×10^{-4}	8.12×10^{-5}	–
8.00×10^{-5}	2.05×10^{-5}	1.99
4.00×10^{-5}	5.13×10^{-6}	1.99
2.00×10^{-5}	1.29×10^{-6}	2.00
1.00×10^{-5}	3.21×10^{-7}	2.00

Table S7: Spinodal decomposition: conservation, energy and cost.

quantity	2D spinodal	3D spinodal
grid	256^2	96^3
ε	0.01	0.02
time steps	5000+4500	1500
initial mass M_h^0	0.1	0.1
largest mass deviation $\lvert M_h^n - M_h^0\rvert$	9.71×10^{-17}	0
largest step change of the modified energy	-3.31×10^{-8}	-4.79×10^{-9}
largest change of E_h between samples	-8.47×10^{-7}	-2.43×10^{-7}
range of $\xi = r/\sqrt{E_1 + C}$	1.00000–1.00035	–
CPU time per step (one core)	1.9 ms	63 ms

Table S8: Maximum nodal error and observed convergence rates for two model problems.

(a) Prescribed values, $-\Delta u = f$, operator D_2

	uniform grid		Chebyshev-graded grid	
N	error	rate	error	rate
8	6.50×10^{-3}	–	3.96×10^{-3}	–
16	1.66×10^{-3}	1.97	1.15×10^{-3}	1.79
32	4.18×10^{-4}	1.99	2.96×10^{-4}	1.95
64	1.05×10^{-4}	2.00	7.46×10^{-5}	1.99
128	2.61×10^{-5}	2.00	1.87×10^{-5}	2.00
256	6.54×10^{-6}	2.00	4.67×10^{-6}	2.00
512	1.63×10^{-6}	2.00	1.17×10^{-6}	2.00

(b) Zero-slope conditions, $(I - \Delta)u = f$, uniform grid

	D_2		$D_2^{(4)}$	
N	error	rate	error	rate
8	3.05×10^{-2}	–	4.69×10^{-3}	–
16	9.11×10^{-3}	1.74	3.15×10^{-4}	3.89
32	2.38×10^{-3}	1.94	2.01×10^{-5}	3.98
64	6.01×10^{-4}	1.98	1.26×10^{-6}	3.99
128	1.51×10^{-4}	2.00	7.87×10^{-8}	4.00
256	3.77×10^{-5}	2.00	4.92×10^{-9}	4.00
512	9.42×10^{-6}	2.00	3.08×10^{-10}	4.00

Table S9: Smooth solution with prescribed displacements on all edges.

N	unknowns	e_{H^1}	rate	e_∞	GMRES its
8	128	9.22×10^{-3}	–	4.13×10^{-3}	15
16	512	2.31×10^{-3}	2.00	1.03×10^{-3}	17
32	2048	5.78×10^{-4}	2.00	2.55×10^{-4}	17
64	8192	1.45×10^{-4}	2.00	6.35×10^{-5}	17
128	32768	3.61×10^{-5}	2.00	1.58×10^{-5}	16
256	131072	9.04×10^{-6}	2.00	3.95×10^{-6}	15

Table S10: Traction boundaries, smooth solution: relative H^1 error for four configurations, GMRES iterations, and maximum displacement error on the traction edges.

	traction on T		traction on T+R	
N	e_{H^1}	rate	e_{H^1}	rate
16	3.30×10^{-3}	–	4.31×10^{-3}	–
32	8.20×10^{-4}	2.01	1.08×10^{-3}	2.00
64	2.05×10^{-4}	2.00	2.70×10^{-4}	2.00
128	5.11×10^{-5}	2.00	6.75×10^{-5}	2.00
256	1.28×10^{-5}	2.00	1.69×10^{-5}	2.00
GMRES its	18–20		27–27	
edge error, N=256	8.64×10^{-6}		1.84×10^{-5}	

	traction on L+B		traction on T+B	
N	e_{H^1}	rate	e_{H^1}	rate
16	3.89×10^{-3}	–	5.99×10^{-3}	–
32	9.70×10^{-4}	2.00	1.49×10^{-3}	2.01
64	2.42×10^{-4}	2.00	3.71×10^{-4}	2.00
128	6.05×10^{-5}	2.00	9.27×10^{-5}	2.00
256	1.51×10^{-5}	2.00	2.32×10^{-5}	2.00
GMRES its	26–27		18–20	
edge error, N=256	1.09×10^{-5}		1.53×10^{-5}	

Table S11: Traction on two adjacent edges, smooth solution, on graded grids.

(a) Chebyshev grid

N	e_{H^1}	rate	edge error	GMRES its
16	6.73×10^{-3}	–	8.10×10^{-3}	26
32	1.69×10^{-3}	1.99	2.04×10^{-3}	26
64	4.23×10^{-4}	2.00	5.10×10^{-4}	26
128	1.06×10^{-4}	2.00	1.27×10^{-4}	26

(b) Geometric grid graded toward the traction edge

N	e_{H^1}	rate	$h_{\max}/h_{\min}$	GMRES its
16	2.79×10^{-2}	–	66	39
32	7.55×10^{-3}	1.89	76	50
64	1.93×10^{-3}	1.97	82	69
128	4.87×10^{-4}	1.99	85	139

Table S12: Corner singularity with prescribed displacements: uniform versus automatic refinement.

(a) Uniform refinement

unknowns	e_{H^1}	GMRES its
128	9.62×10^{-2}	14
512	6.58×10^{-2}	16
2048	4.51×10^{-2}	17
8192	3.09×10^{-2}	16
32768	2.12×10^{-2}	16
131072	1.45×10^{-2}	15

(b) Automatic refinement (Haar-detail indicator)

cycle	unknowns	e_{H^1}	$h_{\min}$	$\kappa(V)$	GMRES its
0	128	9.62×10^{-2}	1.2×10^{-1}	1.0	14
2	220	6.46×10^{-2}	3.1×10^{-2}	2.0	15
4	312	4.45×10^{-2}	1.6×10^{-2}	2.5	15
6	480	3.20×10^{-2}	7.8×10^{-3}	2.9	15
8	756	1.66×10^{-2}	2.0×10^{-3}	3.2	14
10	1456	8.07×10^{-3}	4.9×10^{-4}	3.3	13
12	3040	3.93×10^{-3}	1.2×10^{-4}	3.8	13
14	6380	1.91×10^{-3}	3.1×10^{-5}	5.2	12
16	13608	9.07×10^{-4}	7.6×10^{-6}	5.3	11
18	29280	4.37×10^{-4}	1.9×10^{-6}	7.1	10
20	62300	2.09×10^{-4}	4.8×10^{-7}	7.1	10
22	130048	1.20×10^{-4}	1.2×10^{-7}	9.5	9

Table S13: Allen–Cahn shrinking circle ($\varepsilon = 0.02$, $\Delta t = 0.05$): fitted slope of $R^2(t)$, deviation from $-2\varepsilon^2$, and the largest step changes of the modified and the original energy.

grid	operator	$\mathrm{d}R^2/\mathrm{d}t$	deviation	max $\Delta\tilde{E}$	max ΔE_h
64^2	D_2	-8.138×10^{-4}	1.72%	-8.09×10^{-6}	-1.62×10^{-4}
64^2	$D_2^{(4)}$	-8.008×10^{-4}	0.11%	-7.91×10^{-6}	-1.58×10^{-4}
128^2	$D_2^{(4)}$	-8.028×10^{-4}	0.36%	-7.92×10^{-6}	-1.58×10^{-4}
256^2	$D_2^{(4)}$	-8.030×10^{-4}	0.37%	-7.92×10^{-6}	-1.58×10^{-4}

Table S14: Cahn–Hilliard: spatial self-convergence in the maximum norm.

	D_2		$D_2^{(4)}$	
N	error	rate	error	rate
18	7.95×10^{-4}	–	1.36×10^{-4}	–
54	9.32×10^{-5}	1.95	1.86×10^{-6}	3.91
162	9.40×10^{-6}	2.09	2.24×10^{-8}	4.02

Table S15: Uniform-grid operators within the identical SAV-BDF2 solver.

(a) Smooth Cahn–Hilliard, max error against a pseudo-spectral reference ($N = 486$)

scheme	$N = 18$	$N = 54$	$N = 162$
3-point FD	1.63×10^{-3}	1.89×10^{-4}	2.12×10^{-5}
IHWC, D_2	7.96×10^{-4}	9.44×10^{-5}	1.06×10^{-5}
IHWC, $D_2^{(4)}$	1.36×10^{-4}	1.86×10^{-6}	2.27×10^{-8}
compact (Padé)	9.42×10^{-5}	1.31×10^{-6}	1.60×10^{-8}
pseudo-spectral	4.40×10^{-7}	3.44×10^{-15}	1.61×10^{-15}

(b) Allen–Cahn star, $\varepsilon = 0.004$, against the IHWC $D_2^{(4)}$ reference (1024^2)

	max nodal error		area error	
scheme	256^2	512^2	256^2	512^2
3-point FD	3.89×10^{-1}	8.85×10^{-2}	6.02×10^{-4}	1.46×10^{-4}
IHWC, D_2	1.36×10^{-1}	4.12×10^{-2}	2.43×10^{-4}	6.88×10^{-5}
IHWC, $D_2^{(4)}$	5.27×10^{-2}	2.63×10^{-3}	6.07×10^{-5}	4.23×10^{-6}
compact (Padé)	3.65×10^{-2}	1.78×10^{-3}	3.77×10^{-5}	2.88×10^{-6}
pseudo-spectral	1.52×10^{-2}	1.68×10^{-4}	1.24×10^{-5}	2.72×10^{-7}

Table S16: Fast diagonalization (FDM) versus Bartels–Stewart (B–S) for $D_2U + UD_2^{\mathsf{T}} = F$ on the worst-conditioned of 20 random graded grids per row.

ratio	N	$h_{\max}/h_{\min}$	$\kappa(V)$	backward error FDM	backward error B–S	solution difference
1.5	32	1.1×10^{1}	3.0	1.4×10^{-16}	1.4×10^{-16}	3.5×10^{-15}
1.5	64	4.5×10^{1}	4.8	2.1×10^{-17}	1.9×10^{-17}	2.5×10^{-15}
1.5	128	5.9×10^{2}	10.2	1.4×10^{-17}	7.2×10^{-18}	7.8×10^{-15}
1.5	256	2.0×10^{3}	16.4	1.1×10^{-17}	6.8×10^{-18}	5.4×10^{-13}
2	32	8.3×10^{1}	5.5	3.2×10^{-17}	3.6×10^{-17}	9.9×10^{-16}
2	64	3.3×10^{2}	8.3	2.2×10^{-17}	2.3×10^{-17}	3.3×10^{-15}
2	128	7.5×10^{3}	16.6	4.0×10^{-17}	4.1×10^{-17}	1.8×10^{-15}
2	256	1.4×10^{4}	27.6	4.8×10^{-17}	2.3×10^{-17}	1.5×10^{-10}
4	32	2.5×10^{4}	10.2	9.8×10^{-18}	8.8×10^{-18}	2.5×10^{-13}
4	64	4.2×10^{3}	12.1	6.1×10^{-17}	9.0×10^{-17}	3.0×10^{-10}
4	128	1.1×10^{3}	20.0	2.1×10^{-16}	2.2×10^{-16}	8.0×10^{-10}
4	256	1.2×10^{6}	46.4	5.0×10^{-18}	5.4×10^{-18}	9.6×10^{-8}

Table S17: Refinement indicator versus a true-error oracle for six configurations.

(a) Final relative H^1 error and fitted rate (about 6×10^4 unknowns)

case	λ	uniform 128^2	indicator: error, rate		oracle: error, rate	
face, $\omega = 1.25\pi$	0.674	1.06×10^{-2}	7.85×10^{-5}	-1.05	4.37×10^{-5}	-1.00
face, $\omega = 1.50\pi$	0.544	2.72×10^{-2}	2.47×10^{-4}	-0.98	1.04×10^{-4}	-0.98
face, $\omega = 1.75\pi$	0.505	3.17×10^{-2}	2.97×10^{-4}	-0.96	1.27×10^{-4}	-0.99
face, $\omega = 1.95\pi$	0.500	2.97×10^{-2}	2.18×10^{-4}	-0.96	1.25×10^{-4}	-0.98
two singularities	0.544	4.11×10^{-2}	5.44×10^{-4}	-0.95	3.85×10^{-4}	-0.92
boundary layer	–	3.23×10^{-2}	6.85×10^{-5}	-1.00	2.83×10^{-4}	-0.87

(b) Indicator effectivity, number of refinement cycles, and total setup and solve time

case	effectivity range η/e	cycles	final unknowns	total time (s)
face, $\omega = 1.25\pi$	2.00–5.83	18	70488	0.7
face, $\omega = 1.50\pi$	1.65–4.04	20	71392	0.8
face, $\omega = 1.75\pi$	1.84–3.88	22	69190	0.8
face, $\omega = 1.95\pi$	1.92–5.05	23	69190	0.8
two singularities	1.18–4.05	20	80520	1.0
boundary layer	0.25–15.64	23	81608	0.7

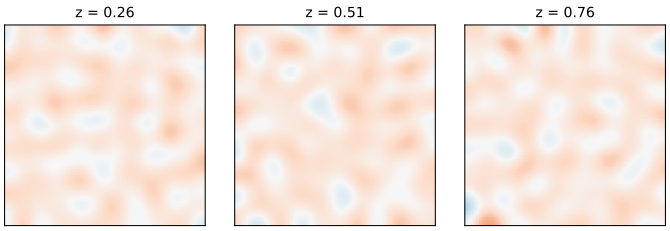


Figure S1: Three-dimensional Cahn–Hilliard on 96^3: slices of the concentration at the final time.

References

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Table S18: Sensitivity to the stabilization S and the time step: largest $|u_h|$, error against the reference at $T = 2000\varepsilon^2$, and ξ at T ($\varepsilon = 0.004$, 384^2).

$\Delta t/\varepsilon^2$	S	$\max_n \lvert u_h \rvert$	error vs. reference	ξ at T
1	0	32.723	1.52×10^{1}	-0.00060
1	1	1.042	1.99×10^{-3}	1.00000
1	2	1.042	3.75×10^{-3}	1.00000
1	4	1.042	7.22×10^{-3}	1.00000
1	8	1.042	1.40×10^{-2}	1.00000
8	0	50.057	2.88×10^{1}	-0.00016
8	1	1.085	1.83×10^{0}	0.87732
8	2	1.042	2.17×10^{-1}	1.00002
8	4	1.047	3.49×10^{-1}	1.00003
8	8	1.234	1.73×10^{0}	1.00002
32	0	31.405	2.53×10^{1}	-0.00022
32	1	1.126	2.06×10^{0}	0.78254
32	2	1.036	2.03×10^{0}	1.00015
32	4	1.069	2.03×10^{0}	1.00025
32	8	1.289	2.03×10^{0}	1.00019

Table S19: Three-dimensional manufactured problem $(I - \Delta)u = f$: maximum nodal error, observed rate and solve time.

(a) Uniform grids

N	unknowns	max error	rate	solve (s)
8	512	9.40×10^{-3}	–	0.0001
16	4096	2.60×10^{-3}	1.86	0.0001
32	32768	6.72×10^{-4}	1.95	0.0007
64	262144	1.69×10^{-4}	1.99	0.0063
96	884736	7.53×10^{-5}	2.00	0.0293

(b) Chebyshev-graded grids

N	unknowns	max error	rate	solve (s)
8	512	1.25×10^{-2}	–	0.0001
16	4096	4.34×10^{-3}	1.53	0.0003
32	32768	1.21×10^{-3}	1.85	0.0014
64	262144	3.10×10^{-4}	1.96	0.0070
96	884736	1.38×10^{-4}	1.99	0.0306